FrostMind — One-Page Summary

1. Theme

Deep Tech & Industry: T-5. Open Source & Innovation

2. Problem Statement

- **Final Version:** Urban working families and multi-generational Indian households struggle with high perishable food wastage (losing ₹1,000–₹2,500 monthly in spoiled dairy, cooked leftovers, and produce) because unorganized storage hides items past expiry, while elderly members frequently miss meal-dependent daily prescriptions.
- **Evolution:** Shifted from a broad, generic complaint about food rotting in fridges to a precise target focusing on household monetary loss, multi-generational mealtime medication routines, and physical storage visibility.

3. What We Built

The Core Flow

1. **Log:** User pastes or scans raw receipt text / barcode input into the Streamlit dashboard.
2. **Process & Store:** A local Python parser cleans text using fuzzy matching (`difflib`), auto-assigns default shelf-life expiration dates, and saves items to a local SQLite database (`inventory.db`).
3. **Trigger:** A physical door-open event fires during a designated meal window (e.g., Breakfast).
4. **Display:** The system renders a single ambient card displaying expiring food items alongside scheduled meal-dependent medication reminders.

Explicit Scope Boundaries (What it DOES NOT do)

- **No Cloud Accounts:** Strictly single-household, zero multi-tenant cloud authentication.
- **100% Offline:** Zero external third-party cloud API dependencies or LLM network calls.
- **No Video Decay Modeling:** Relies strictly on receipt/barcode logging and database timers rather than real-time camera computer vision.
- **Focused Domain:** No payment processing, automated e-commerce reordering, or non-kitchen object tracking.

4. Tech Stack

Layer	Technology	Selection Rationale
Frontend / UI	**Streamlit**	Rapid, pure-Python dashboard rendering optimized for mounted touch-screen displays.
Backend Logic	**Python 3.11 + `difflib` + Regex**	Lightweight, fast text normalization, fuzzy string matching, and offline receipt parsing without heavy dependencies.
Data & Storage	**SQLite3**	Zero-configuration, serverless, single-file relational database running 100% locally.
Target Hardware	**Raspberry Pi Zero 2 W / Laptop + Reed Switch + E-ink**	Low power consumption, total system cost under ₹4,000, and complete hardware-level network isolation.

5. Evidence Position

- **Proven Facts:**
  - Household food waste is a massive, quantifiable global issue (631M tonnes/year, UNEP Food Waste Index).
  - Thermal receipt OCR text can be normalized offline using `difflib` fuzzy matching to correct character errors and strip non-food items.
  - Habit-anchoring reminders to physical door-open events during meal times improves visibility compared to standalone mobile alarms.
- **Untested Assumptions:**
  - Long-term user discipline: Whether users will consistently scan receipts or barcodes after every grocery trip over several months without abandoning the logging habit.

6. What We Would Build Next

- **The Next Flow:** On-device camera-based barcode and Tesseract OCR scanning combined with local regional language voice prompts (Hindi/Tamil/Telugu).

- **Validation Requirement:** Field testing must prove that users find taking 5 seconds to scan a receipt significantly lower friction than manual entry, and that non-technical household members respond better to ambient voice alerts than digital screen cards.